Related Work: Graph-Based Intrusion Detection on UNSW-NB15

CSE475 Machine Learning · Task 1 · Group 7 · Track 1 (GNN) · East West University

Five peer-reviewed papers, 2022–2026, each verified against its full text.

1. Overview

Paper	Title	Authors	Year	GNN type	Headline UNSW-NB15 result
E-GraphSAGE	E-GraphSAGE: A Graph Neural Network based Intrusion Detection System for IoT	Wai Weng Lo, Siamak Layeghy, Mohanad Sarhan, Marcus Gallagher, Marius Portmann	2022	E-GraphSAGE	Accuracy 99.35%; Precision 0.97
FN-GNN	FN-GNN: A Novel Graph Embedding Approach for Enhancing Graph Neural Networks in Network Intrusion Detection Systems	Dinh-Hau Tran, Minho Park	2024	Hybrid GCN + SAGEConv	Accuracy 98.65%; Recall 98.38–99.51%*
DIGNN-A	DIGNN-A: Real-Time Network Intrusion Detection with Integrated Neural Networks Based on Dynamic Graph	Jizhao Liu, Minghao Guo	2025	Dynamic GNN	Accuracy 99.3%; F1 97.22%
GCN-KNN	Enhancing network intrusion detection using graph convolutional networks fused with traditional systems	Patrick Ansah, Caroline John, Sumit Kumar Tetarave, Ezhil Kalaimannan	2025	GCN fused with k-nearest neighbours	Accuracy 99.94%; Precision 99.90%
Nwachukwu	Rethinking Graph-Based Approaches: An Empirical Study of Feature Engineering in Network Intrusion Detection	Richard Chukwuebuka Nwachukwu	2026	None. Graph-theoretic features	Accuracy 0.934; Precision 0.954

2. How each paper evaluated: the source of our research gap

This table compares evaluation conditions rather than scores. Red cells mark a condition that inflates a reported result or limits its comparability; green cells mark good practice.

Paper	Data used	Split	Validation set	Class distribution	Headline metric	Topology
E-GraphSAGE	Original, 2.54M flows	Random 70/30	None	Natural	Weighted F1	Real IP:port
FN-GNN	Original, 50k subsample	Random 80/20	Yes	Rebalanced 61/39	Weighted F1	Real IPs
DIGNN-A	Original, ~2.6M	0.3 ratio, unspecified	None stated	Natural	F1 (binary)	Real IP:port
GCN-KNN	Original, 44k subset	Not stated	Not stated	Rebalanced 50/50	Accuracy, F1	Not described
Nwachukwu	Original features	Not stated	Not stated	Natural	Accuracy, F1	SYNTHETIC IPs
Our project	Original, full release	Host-disjoint 70/15/15	Yes, mandatory	Natural	Macro-F1 + weighted F1	Real IPs

Reading: no paper splits by host. Two of the four GNN papers report near-ceiling scores on artificially rebalanced subsets. The only paper that tests whether graph structure helps used a synthetic topology. Section 4 states the gap these conditions leave.

3. Paper-by-paper review

3.1 E-GraphSAGE

Title	**E-GraphSAGE: A Graph Neural Network based Intrusion Detection System for IoT**				
Authors	Wai Weng Lo, Siamak Layeghy, Mohanad Sarhan, Marcus Gallagher, Marius Portmann (University of Queensland)				
Year / venue	IEEE/IFIP Network Operations and Management Symposium (NOMS), 2022				
Link	arxiv.org/abs/2103.16329				
Dataset	UNSW-NB15, original release (49 features, 2,540,044 flows). Also NF-UNSW-NB15 (NetFlow), BoT-IoT, ToN-IoT and their NF- versions.				
Application area	Network intrusion detection for IoT networks				
GNN type	E-GraphSAGE: GraphSAGE extended to aggregate edge features (inductive, edge classification)				
Graph construction	Nodes = network endpoints (IP:port). Edges = flows, carrying the flow features. Node features initialised to a constant vector of ones.				
Method / model	Message passing aggregates the features of the edges incident to each node's sampled neighbourhood. A flow is classified from the concatenated embeddings of its two endpoint nodes, followed by a softmax layer.				
Split	70 / 30 train-test on the full dataset. No stratification, seed or temporal ordering stated. No validation set.				
Metrics reported	**Accuracy**	**Precision**	**Recall**	**F1**	**Other**
Binary	99.35%	0.97	97.51%	0.97	FAR 0.38%
Multiclass	98.15%	—	—	Weighted F1 0.98	—
Strengths	• First GNN-based NIDS to use edge features directly, which matches the natural structure of flow data. • Inductive: classifies flows involving hosts unseen in training, which is required for a host-disjoint evaluation. • Evaluated on six datasets; public reference implementations exist. • Reports on the original UNSW-NB15 release, so its numbers are directly comparable to ours.				
Limitations	• No validation set: hyperparameter and checkpoint selection are not separated from the test data. • Random split: flows from the same host fall on both sides, so the model can partly recognise hosts rather than behaviour. • Multiclass result reported as weighted F1 only, which is dominated by the majority classes on data where Worms has 44 samples. • On UNSW-NB15 multiclass it is outperformed by a non-graph CNN-WDLSTM (98.43% accuracy) in its own comparison table.				
Research gap	**Performance under host-disjoint evaluation, and macro-F1 across all ten classes, are not reported.**				
Relation to our work	Primary reference. We adopt its flow-as-edge formulation and inductive design. Its original-variant numbers are our main Pillar A comparison target. We differ in using a three-way host-disjoint split and in reporting macro-F1 alongside weighted F1.				
Comparable to our results?	**Yes**				

3.2 FN-GNN

Title	**FN-GNN: A Novel Graph Embedding Approach for Enhancing Graph Neural Networks in Network Intrusion Detection Systems**				
Authors	Dinh-Hau Tran, Minho Park (Soongsil University, Seoul)				
Year / venue	Applied Sciences, vol. 14, no. 16, art. 6932, 2024 (open access)				
Link	mdpi.com/2076-3417/14/16/6932				
Dataset	UNSW-NB15, original release (49 features). Subsample of 50,000 flows. Also CICIDS-2017.				
Application area	Network intrusion detection				
GNN type	Hybrid GCN + SAGEConv (GraphSAGE convolution)				
Graph construction	Nodes initialised with values derived from flow features. Edges built by analysing IP relationships within the network.				
Method / model	A preprocessing pipeline standardises flow data into graph input; GCN and SAGEConv layers are combined to learn node representations for malicious-flow detection.				
Split	80 / 20 random split; 20% of the training portion held out for validation. Test set: 9,997 flows.				
Metrics reported	**Accuracy**	**Precision**	**Recall**	**F1**	**Other**
Binary	98.65%	—	98.38–99.51%*	Weighted F1 98.65%	—

** Range reported across both datasets combined.*

Strengths	• Directly tackles the lack of standardised graph input for NIDS. • Uses a validation set, unlike E-GraphSAGE. • Uses the original UNSW-NB15 release. Open access, so fully checkable.
Limitations	• Subsample deliberately rebalanced to 30,614 normal (61.22%) and 19,386 attack (38.78%) flows, against a real distribution of roughly 87% / 13%. • Only 32 of the 49 features retained. • Random split; weighted F1 only.
Research gap	**Its score is measured on an artificially balanced subsample, so it does not show performance on the real class distribution.**
Relation to our work	Comparison target, qualified: our evaluation keeps the natural distribution, so a lower macro-F1 from us is not evidence of a weaker model. Its rebalancing is direct evidence for our research gap.
Comparable to our results?	**Yes, with caveat**

3.3 DIGNN-A

Title	**DIGNN-A: Real-Time Network Intrusion Detection with Integrated Neural Networks Based on Dynamic Graph**				
Authors	Jizhao Liu, Minghao Guo (Zhongyuan University of Technology, Zhengzhou)				
Year / venue	Computers, Materials & Continua, vol. 82, no. 1, pp. 817–842, 2025				
Link	doi.org/10.32604/cmc.2024.057660				
Dataset	UNSW-NB15, original release (approx. 2.6 million records, 49 features). Also NF-ToN-IoT-v2.				
Application area	Real-time network intrusion detection				
GNN type	Dynamic GNN: 3-layer TransformerConv (spatial) + 2-layer GCNII-GRU (temporal), on line graphs				
Graph construction	Nodes = sockets (IP:port). Edges = communications, with protocol, state and byte size as features. Traffic sorted by timestamp and cut into fixed-size snapshots; each snapshot converted to a line graph so that edge classification becomes node classification.				
Method / model	Graph attention extracts structure within each snapshot; a GRU cell built on GCNII carries information across a window of consecutive snapshots. Best batch size 2048, window size 10.				
Split	Train-test ratio 0.3. Stratification and whether the split respects time order are not stated.				
Metrics reported	**Accuracy**	**Precision**	**Recall**	**F1**	**Other**
Binary	99.3%	—	—	97.22%	FAR 0.03

Multiclass results are not reported.

Strengths	• Models how network topology changes over time, which a static graph cannot. • Line-graph transformation is an elegant alternative to native edge classification. • Reports false-alarm rate, which matters operationally.
Limitations	• Binary classification only; no multiclass or per-class results. • Split procedure underspecified despite the temporal design. • Authors acknowledge the model is hard to interpret, and that line graphs collapse to isolated points when traffic forms independent pairs. • FAR on UNSW-NB15 is higher than the CTGNN baseline (0.0199), which the authors attribute to imbalance.
Research gap	**Multiclass and minority-class performance, and evaluation on hosts unseen in training, are not reported.**
Relation to our work	Methodological contrast for our ablation: line-graph node classification versus E-GraphSAGE's direct edge classification. Its 99.3% binary accuracy is a comparison target for our binary results.
Comparable to our results?	**Yes (binary only)**

3.4 GCN-KNN

Title	**Enhancing network intrusion detection using graph convolutional networks fused with traditional systems**				
Authors	Patrick Ansah, Caroline John, Sumit Kumar Tetarave, Ezhil Kalaimannan (University of West Florida)				
Year / venue	International Journal of Data Science and Analytics, vol. 21, no. 1, art. 31, Springer, 2025				
Link	doi.org/10.1007/s41060-025-00922-9				
Dataset	UNSW-NB15, original release (approx. 700,000 instances, 49 features). Balanced subset of 22,000 anomalous + 22,000 normal records (44,000 total).				
Application area	Proactive anomaly detection and intrusion classification				
GNN type	GCN fused with k-nearest neighbours (GCN-KNN, supervised); GCN autoencoder (GCN-AE, unsupervised)				
Graph construction	Graph structures encode node-to-node relationships; construction details are not given in the abstract.				
Method / model	Two models: a GCN autoencoder for unsupervised anomaly detection, and a GCN whose embeddings feed a k-NN classifier for supervised detection. Keras/TensorFlow with hyperparameter optimisation.				
Split	Not stated.				
Metrics reported	**Accuracy**	**Precision**	**Recall**	**F1**	**Other**
GCN-KNN	99.94%	99.90%	99.97%	99.93%	AUC 0.99
GCN-AE	52.86%	81.48%	7.96%	14.51%	—
Strengths	• Compares supervised and unsupervised graph models on the same data. • Fuses a graph encoder with a classical classifier, close in spirit to our baseline-versus-GNN comparison. • Uses the original UNSW-NB15 release.				
Limitations	• Balanced 22,000 / 22,000 subset bears no relation to the real class distribution. • Split method not stated, so the near-perfect 99.94% cannot be checked for leakage. • The unsupervised GCN-AE recovers only 7.96% of attacks, which the summary does not discuss. • Binary only; no per-class results.				
Research gap	**Performance on the natural, imbalanced distribution and on unseen hosts is unknown.**				
Relation to our work	Second example (with FN-GNN) of a near-ceiling score obtained on an artificially balanced subsample. Supports our decision to keep the natural distribution and judge on macro-F1.				
Comparable to our results?	**Yes, with caveat**				

3.5 Nwachukwu

Title	**Rethinking Graph-Based Approaches: An Empirical Study of Feature Engineering in Network Intrusion Detection**				
Authors	Richard Chukwuebuka Nwachukwu (Ignatius Ajuru University of Education, Port Harcourt)				
Year / venue	Jurnal Teknologi Informatika (JTIF), vol. 14, no. 1, pp. 1–10, 2026				
Link	doi.org/10.21063/jtif.2026.V14.1.1-10				
Dataset	UNSW-NB15 (original feature set; class distribution retained). Exact record count not reported.				
Application area	Network intrusion detection: methodological evaluation of graph features				
GNN type	None. Graph-theoretic features (degree, betweenness, closeness, eigenvector centrality, PageRank, Louvain communities) fed to classical classifiers				
Graph construction	SYNTHETIC topology: real IPs replaced with generated addresses (192.168.x.y internal, 10.a.b.c external) for privacy. 1,999 nodes, 14,736 edges, density 0.0074, diameter 5.				
Method / model	Compares RBF network, Random Forest and RBF-kernel SVM trained on traditional flow features versus the same features enhanced with graph-theoretic features.				
Split	Not specified beyond a general reference to cross-validation.				
Metrics reported	**Accuracy**	**Precision**	**Recall**	**F1**	**Other**
RF, traditional	0.934	0.954	0.925	0.940	—
RF, + graph	0.926	0.955	0.909	0.931	—
RBFN, traditional	0.910	0.952	0.896	0.923	—
RBFN, + graph	0.840	0.859	0.865	0.862	—
SVM, traditional	0.860	0.898	0.842	0.869	—
SVM, + graph	0.842	0.871	0.837	0.853	—

Graph feature extraction took 15.3 s versus 0.8 s for traditional features, a 19.1× overhead.

Strengths	• The only paper in our review that directly tests whether graph information adds value over flat features. • Controlled comparison: same classifiers, with and without graph features. • Reports computational cost, which other papers omit.
Limitations	• The topology is synthetic, not the real host graph; the author states real topologies might yield different results. • No graph neural network evaluated, only hand-computed graph statistics. • Split not specified; significance claimed without a reported test.
Research gap	**Whether the REAL host topology, learned by a GNN rather than summarised as statistics, improves detection.**

Relation to our work	Our sceptical counterweight and the sharpest statement of our gap. It finds graph features hurt, but on a synthetic topology. We test the real topology, recovered from the IP columns, inside a GNN. If our GNN only matches the flat baseline, this paper makes that a finding consistent with recent literature.
Comparable to our results?	**Methodological only**

4. Synthesis

4.1 What the literature shows

- Graph models report high scores on UNSW-NB15: 98.15–99.94% accuracy across the four GNN papers.
- None of the five papers splits data by host, so models can recognise hosts that appear in both training and test sets.
- Two papers (FN-GNN, GCN-KNN) rebalance the data before evaluation, removing the imbalance that makes the dataset hard.
- Three of five report weighted F1 or accuracy as the headline metric; none reports macro-F1 across the ten classes.
- The one controlled test of whether graph information helps (Nwachukwu, 2026) finds it hurts, but on a synthetic topology, and without a GNN.

4.2 Research gap

Published GNN results on UNSW-NB15 are obtained under random splits and, in two cases, on artificially rebalanced subsets. The only study that isolates the contribution of graph structure used a synthetic topology rather than the real one. It is therefore unknown whether the real host topology contained in UNSW-NB15 improves detection when evaluated on hosts unseen during training and on the natural class distribution.

4.3 Our research question

> **Does a host-topology graph improve macro-F1 over a k-nearest-neighbour feature-similarity graph on UNSW-NB15, under the same three-way host-disjoint split and the same training budget?**

4.4 Which results we compare against

Our Pillar A comparison uses papers evaluated on the same original 49-feature release. Because these papers report weighted F1 or binary metrics, we report weighted F1 and binary metrics alongside macro-F1, and always compare like with like. Results from rebalanced subsets (FN-GNN, GCN-KNN) are reported with that qualification stated. Nwachukwu (2026) is a methodological reference only, since it evaluates graph features with classical classifiers rather than a GNN.

4.5 Excluded paper

Anomal-E (Caville, Lo, Layeghy, Portmann; Knowledge-Based Systems, 2022) was reviewed and excluded from the comparison set. It evaluates on NF-UNSW-NB15-v2, a 43-feature NetFlow re-extraction of the same traffic, so its scores (best macro-F1 92.35%) are not comparable with results on the original release. It is cited as a reference for self-supervised graph learning.

References

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